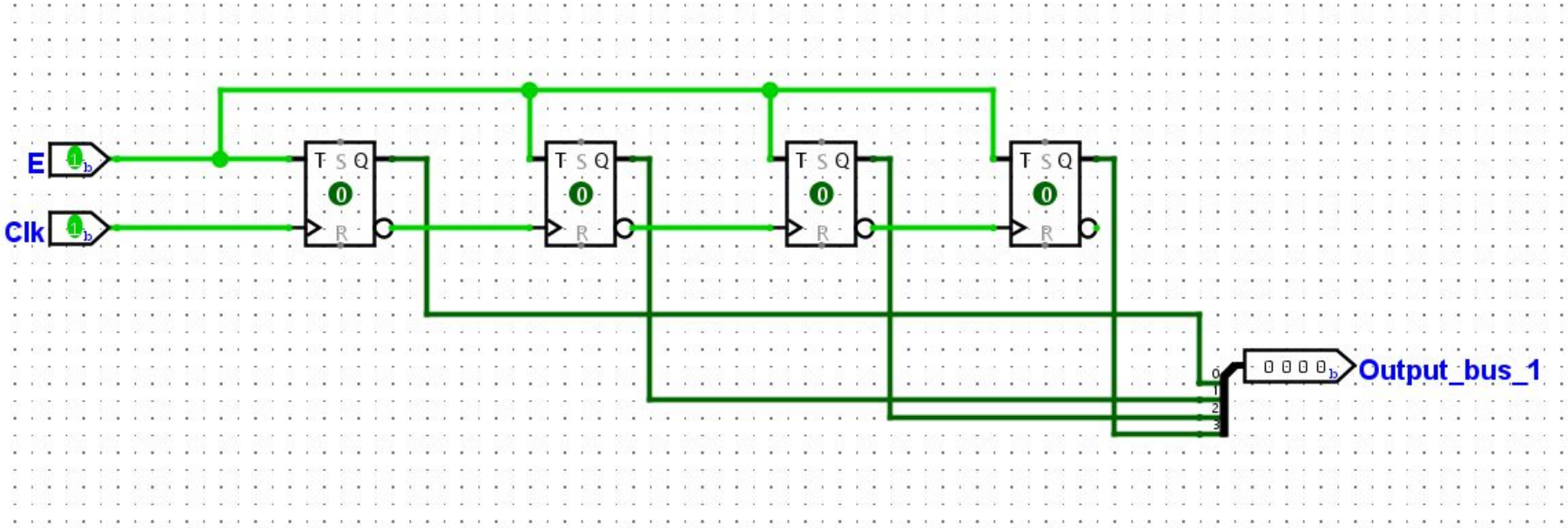
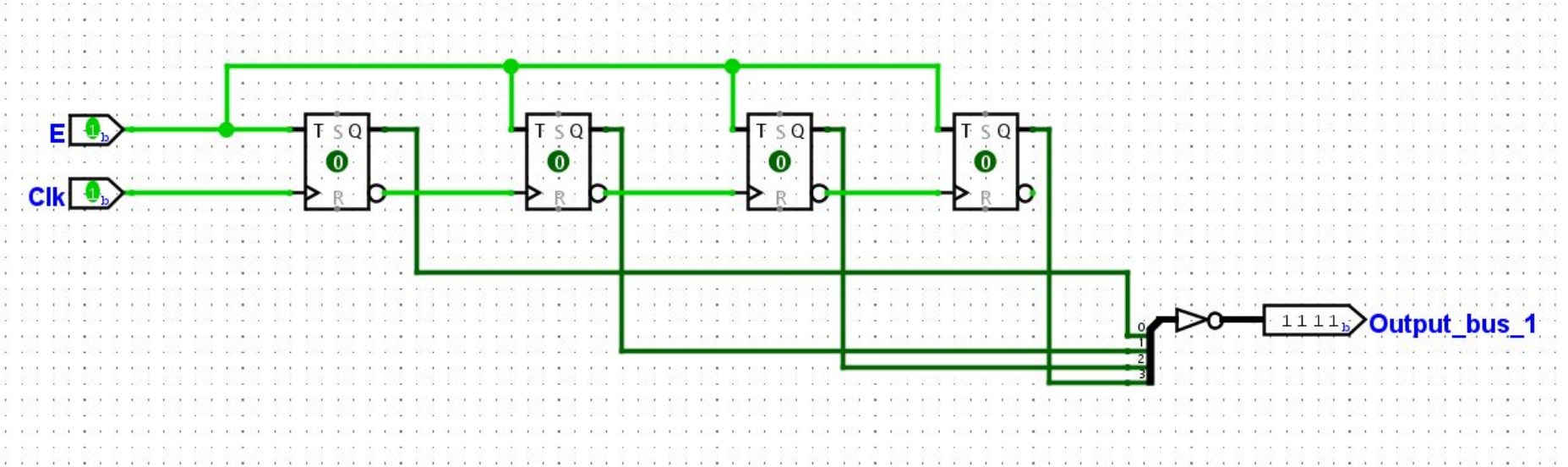
Up-Counter



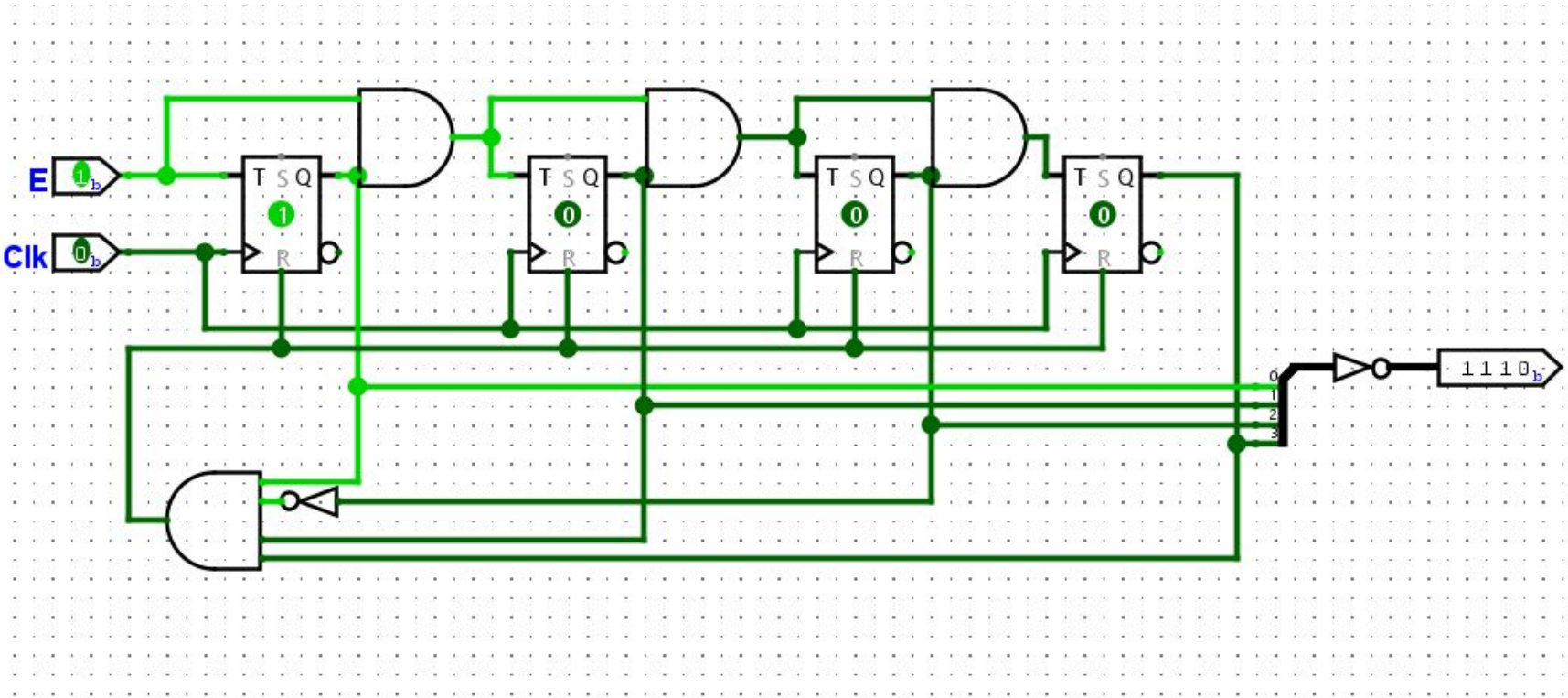
counts up 0-15, need to make this but backwards somehow

Down-Counter

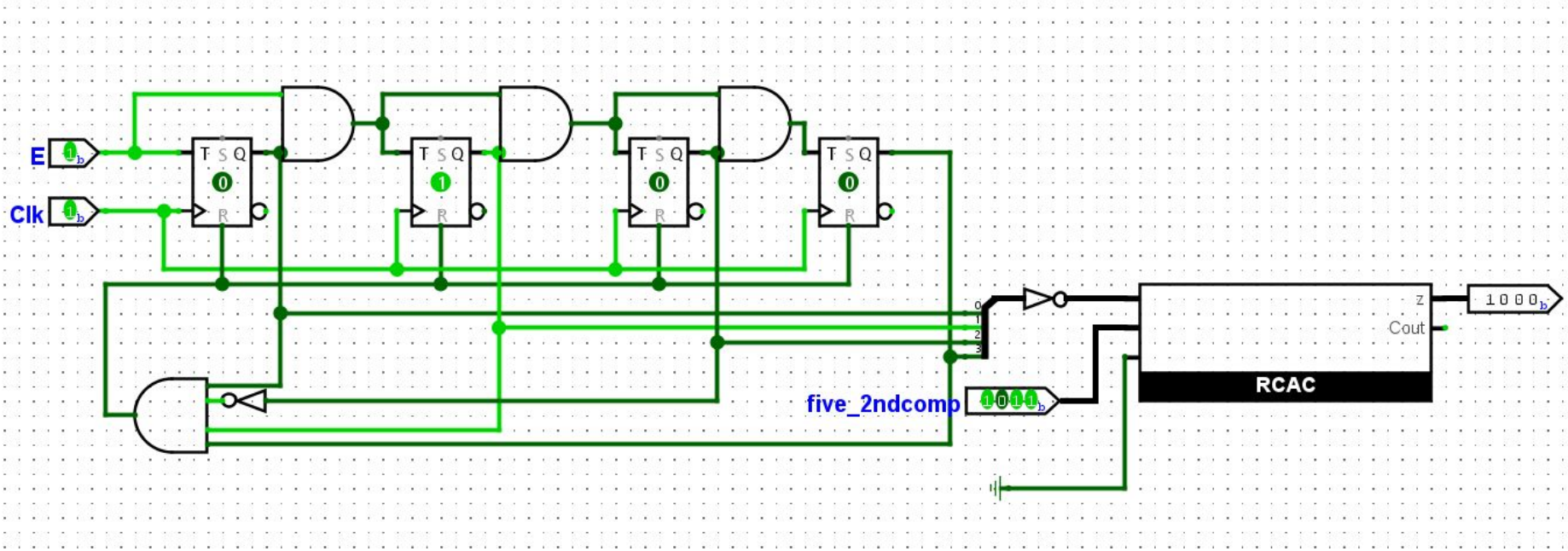


by just negating the output of the up counter, this counts 15-0

Down-Counter Revision

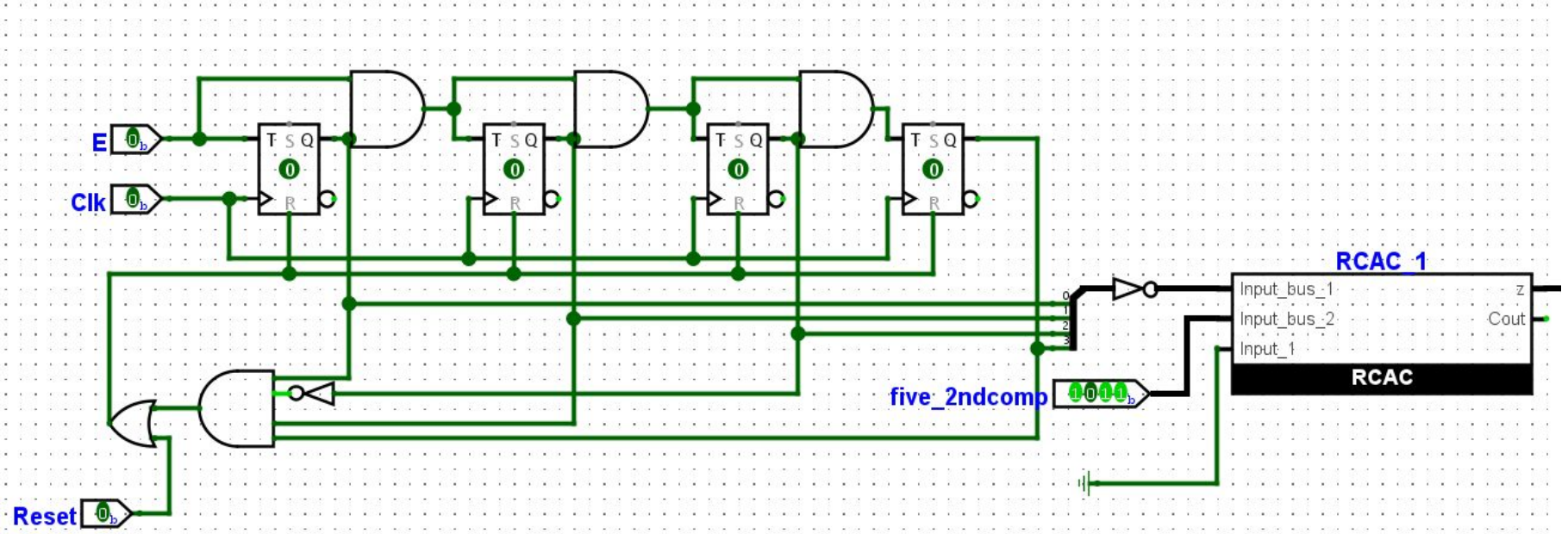


this counts down from 15-5… almost what i want, but not quite
i need to take the output from this and subtract 5 somehow



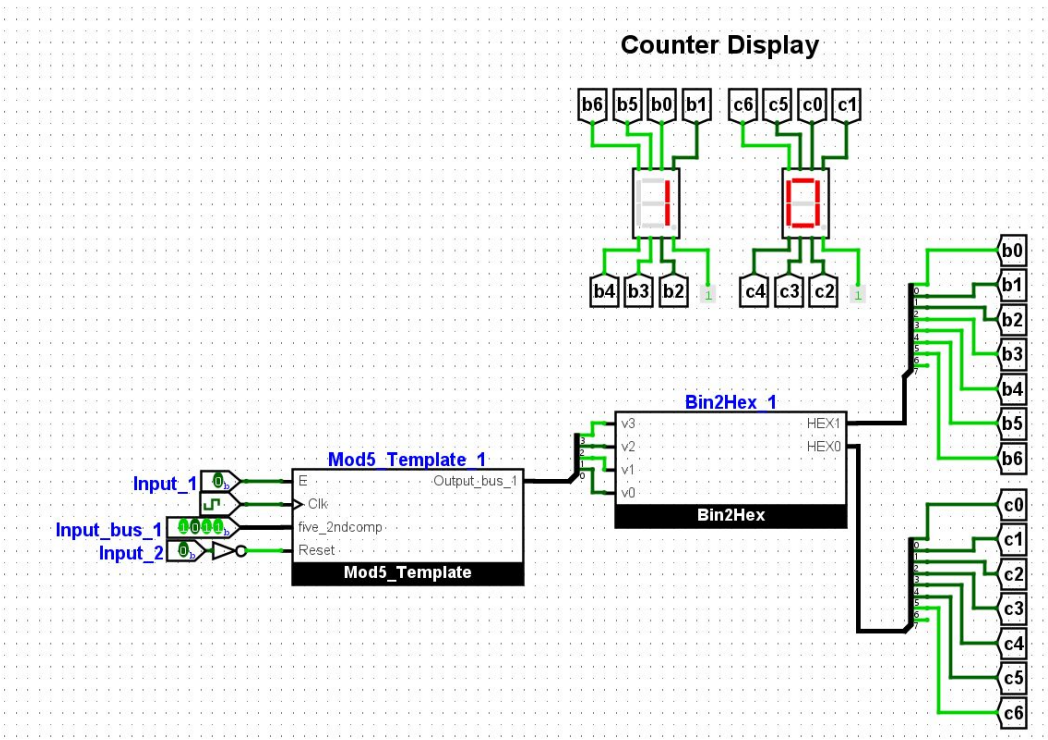
using the ripple carry adder from a previous lab i can add the second inverse of 5 to get the results i want. this successfully counts down from 10-0, even if it is a little dirty

Reset Button



as simple as adding an or gate and an input, now the user can decide when it resets

Integration With 7-Seg Display



the reset input needs to be negated since an unpressed button on the board outputs a 1